ARITHMETIC SHADOWS OF CHIRAL ALGEBRAS

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Abstract. The scalar genus-1 shadow channel of a chiral algebra has Fourier coefficients $-24\kappa(\mathcal{A})\sigma_1(n)$, hence Dirichlet series

$$D_2(\mathcal{A}, s) = -24\kappa(\mathcal{A})\zeta(s)\zeta(s-1).$$

For lattice vertex algebras, the constrained Epstein zeta function on the punctured lattice sector recovers the Riemann zeta function in rank one and, in higher rank, has completed Mellin transform given by the lattice theta series. Ordered chiral homology at integer level recovers the Verlinde formula; the Verlinde dimensions are polynomials whose leading coefficients are normalized even zeta values. At genus 2, the bar-complex theta component connects to Siegel modular forms and to the Furusawa–Morimoto form of Böcherer's theorem.

Contents

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1. Introduction

1.1. The arithmetic content of the shadow tower. The bar complex of a chiral algebra $\mathcal{A}$ on an algebraic curve X is a cochain complex $\bar{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct and differential encoding the OPE through the logarithmic kernel $\eta = d\log(z_1 - z_2)$. Over a family of genus-g curves, the fiberwise differential squares to $d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$, where $\omega_g = c_1(\lambda)$ is the first Chern class of the Hodge bundle on $\overline{\mathcal{M}}_g$. The single scalar $\kappa(\mathcal{A})$, the *modular characteristic*, controls the obstruction to flatness of the bar complex as a family.

The shadow obstruction tower $\{S_r(\mathcal{A})\}_{r\geq 2}$ extracts the degree-r constant terms of the universal Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$, with $S_2 = \kappa$. The shadow coefficients are rational functions of the central charge and level, carrying algebraic data from the OPE into the tower.

These rational numbers carry arithmetic content beyond their algebraic origin.

1.2. The Heisenberg base point. The Heisenberg algebra $\mathcal{H}_k$ is the abelian Kac–Moody algebra at level k, with OPE $J(z)J(w) \sim k/(z-w)^2$ and modular characteristic $\kappa(\mathcal{H}_k) = k$. The genus-1 amplitude of $\mathcal{H}_k$ over the family $C_\tau \to \mathfrak{H}$ is

$$F_1^{\mathrm{conn}}(\mathcal{H}_k; q) = -k \sum_{n\geq 1} \sigma_1(n)\, q^n = k \cdot \frac{E_2^*(\tau) - 1}{24},$$

where $E_2^*(\tau) = 1 - 24\sum_{n\geq 1}\sigma_1(n)\, q^n$ is the quasi-modular Eisenstein series and $\sigma_1(n) = \sum_{d\mid n} d$ is the sum-of-divisors function. The Fourier coefficients are divisor sums: the simplest arithmetic functions in analytic number theory.

The Heisenberg computation is the base model. The affine Kac–Moody, Virasoro, $\mathcal{W}_N$, and lattice families add the Lie-theoretic and arithmetic terms one at a time.

1.3. Summary of results. The main arithmetic surfaces are the Eisenstein factorisation of D_2, the constrained Epstein zeta of lattice vertex algebras, the elementary convergence range for the formal shadow Dirichlet series, the Verlinde polynomial family, and the genus-2 Siegel modular projection problem.

Section 7 connects the genus-2 Maurer–Cartan element to Siegel modular forms via the Saito–Kurokawa lift and the Böcherer theorem.

Section 8 develops the constrained Epstein zeta function, its Hecke decomposition, and the Dirichlet–sewing lift.

1.4. **Conventions.** All chiral algebras are $\mathbb{Z}$-graded with cohomological convention ($|d| = +1$). The modular characteristic is

$$\kappa(\mathcal{A}) := \begin{cases} k & \text{Heisenberg } \mathcal{H}_k, \\ \dfrac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee} & \text{affine Kac–Moody } V_k(\mathfrak{g}), \\ c/2 & \text{Virasoro } \mathrm{Vir}_c, \\ c\,(H_N - 1) & \text{principal } \mathcal{W}_N, \end{cases}$$

where $H_N = \sum_{j=1}^{N} 1/j$ is the N-th harmonic number. We write $q = e^{2\pi i\tau}$ for the nome. The ordered bar complex is $\bar{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal and $|s^{-1}v| = |v| - 1$ is the desuspension.

1.5. **Relation to prior work.** The modular-form techniques used in Sections 2–7 draw on the classical theory of Eisenstein series, Hecke operators, and L-functions as developed in Zagier [13] and the systematic treatment of analytic methods for L-functions and automorphic forms in Iwaniec–Kowalski [7]. The factorisation of the shadow Dirichlet series $D_2(\mathcal{A}, s)$ into products of Riemann zeta functions (Theorem 2.2) is an instance of the Rankin–Selberg method, connecting the arithmetic content of OPE data to classical multiplicative number theory. At higher rank, the constrained Epstein zeta function is the Mellin transform of a theta series and decomposes through the Hecke eigenbasis of the corresponding modular-form space.

2. The shadow Eisenstein theorem

2.1. **Genus-1 amplitudes from the bar complex.** Let $\mathcal{A}$ be a chirally Koszul algebra with modular characteristic $\kappa = \kappa(\mathcal{A})$. In the scalar shadow channel, or in a diagonal Heisenberg/weight-multiset model satisfying the required Hilbert-Schmidt sewing estimates, the genus-1 connected free energy has Fourier coefficients proportional to $\sigma_1(n)$. The full genus-1 determinant requires the separate analytic sewing hypotheses.

The genus-1 shadow amplitude is

$$\mathrm{Sh}_2^{(1)}(\mathcal{A}, \tau) = \kappa(\mathcal{A}) \cdot E_2^*(\tau), \tag{2.1}$$

where $E_2^*(\tau) = 1 - 24\sum_{n\geq 1} \sigma_1(n)\, q^n$ is the quasi-modular Eisenstein series (weight 2, with the non-holomorphic completion $\hat{E}_2(\tau) = E_2^*(\tau) - 3/(\pi\,\mathrm{Im}(\tau))$ absorbing the anomaly from $d\log E(z, w)$). The Fourier coefficients $\sigma_1(n) = \sum_{d|n} d$ are the sum-of-divisors function.

Example 2.1 (Explicit genus-1 amplitudes)**.** For the standard families:

(a) *Heisenberg* $\mathcal{H}_k$*:* $\kappa(\mathcal{H}_k) = k$; $\mathrm{Sh}_2^{(1)} = k \cdot E_2^*(\tau)$.

(b) *Affine Kac–Moody* $V_k(\mathfrak{g})$*:* $\kappa(V_k(\mathfrak{g})) = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$; $\mathrm{Sh}_2^{(1)} = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee) \cdot E_2^*(\tau)$. At $k = 0$, $\kappa = \dim(\mathfrak{g})/2$, so the Heisenberg summand contributes a nonzero amplitude. At $k = -h^\vee$: $\kappa = 0$; the amplitude vanishes (critical level).

(c) *Virasoro* Vir_c*:* $\kappa(\mathrm{Vir}_c) = c/2$; $\mathrm{Sh}_2^{(1)} = (c/2) \cdot E_2^*(\tau)$. Self-dual at $c = 13$, where $\kappa + \kappa' = 13$.

2.2. **The Dirichlet series of divisor sums.** The Fourier coefficients $\sigma_1(n)$ assemble into a Dirichlet series with a classical factorisation.

Theorem 2.2 (Shadow Eisenstein theorem). *Let $\mathcal{A}$ be a chirally Koszul algebra with modular characteristic $\kappa(\mathcal{A})$. Define $a_n(\mathcal{A}) = -24\kappa(\mathcal{A})\sigma_1(n)$. The scaled Dirichlet series*

$$D_2(\mathcal{A}, s) := \sum_{n\geq 1} a_n(\mathcal{A})\, n^{-s}$$

satisfies

$$D_2(\mathcal{A}, s) = -24\kappa(\mathcal{A})\,\zeta(s)\,\zeta(s-1),$$

where ζ is the Riemann zeta function. The genus-1 amplitude Dirichlet series is therefore

$$-24\,\kappa(\mathcal{A})\cdot\zeta(s)\cdot\zeta(s-1):$$

an Eisenstein L-function with the modular characteristic as the sole chiral-algebra input.

Proof. The identity $\sigma_1(n) = \sum_{d|n} d$ is a Dirichlet convolution: $\sigma_1 = \mathrm{id} * \mathbf{1}$, where $\mathrm{id}(n) = n$ and $\mathbf{1}(n) = 1$. The Dirichlet series of a convolution is the product of the constituent Dirichlet series:

$$\sum_{n\geq 1} \sigma_1(n)\, n^{-s} = \Big(\sum_{n\geq 1} n^{1-s}\Big)\Big(\sum_{n\geq 1} n^{-s}\Big) = \zeta(s-1)\,\zeta(s).$$

The factor $-24\,\kappa(\mathcal{A})$ comes from the relation (2.1): the genus-1 amplitude has Fourier coefficient $-24\,\kappa(\mathcal{A})\,\sigma_1(n)$ at q^n. □

Remark 2.3 (Analytic properties). The product $\zeta(s)\,\zeta(s-1)$ has simple poles at $s = 1$ and $s = 2$, a functional equation inherited from the individual zeta factors, and zeros at the nontrivial zeros of ζ shifted by 0 and 1. The residue at $s = 2$ is $\mathrm{Res}_{s=2}\,\zeta(s)\zeta(s-1) = \zeta(2) = \pi^2/6$.

Remark 2.4 (Non-automorphicity of the shadow tower). The shadow L-function $L^{\mathrm{sh}}_{\mathcal{A}}(s) = \sum_{r\geq 2} S_r\, r^{-s}$, built from the shadow tower coefficients S_r, is a different object from $D_2(\mathcal{A}, s)$. No Euler product or Selberg-class structure is part of its definition. Analytic continuation requires an explicit generating-function theorem for the full shadow tower.

2.3. **The Polyakov correspondence.** The shadow Eisenstein theorem has a physical counterpart. Polyakov's determinant calculation computes the string partition function by gauge-fixing the worldsheet metric to conformal gauge, producing a conformal anomaly $\langle T^a{}_a\rangle = (c/12)\,R$ and a ghost system with $c_{\mathrm{ghost}} = -26$. The bar-cobar side records the corresponding shadow coefficients.

Polyakov	**Bar-cobar**
Conformal anomaly $(c/12)\,R$	Modular characteristic $\kappa(\mathcal{A})$
$\det'_\zeta \Delta$ (Laplacian)	Fredholm $\det(1-K)$ (Bergman kernel)
Critical dimension $c = 26$	$\kappa(\mathrm{Vir}_{26-c}) = 0$
Ghost system bc	Koszul dual $\mathcal{A}^!$
$F_g = \kappa\,\lambda_g^{\mathrm{FP}}$ (uniform)	Genus-g shadow amplitude

The genus-1 functional determinant $\det'_\zeta \Delta = |\eta(\tau)|^4\,(\mathrm{Im}\,\tau)^2$ is the Heisenberg contribution; the divisor sums in E_2^* encode the one-loop corrections. The shadow Eisenstein theorem extracts the multiplicative structure of these corrections.

3. The categorical zeta function

3.1. **Lattice vertex algebras and the punctured lattice sector.** Let Λ be an even lattice of rank r with inner product $\langle\cdot,\cdot\rangle$, and let V_Λ be the associated lattice vertex algebra. The vertex algebra V_Λ decomposes as $V_\Lambda = \mathcal{H}^{\otimes r}\otimes\mathbb{C}[\Lambda]$, where $\mathcal{H}^{\otimes r}$ is the rank-r Heisenberg algebra and $\mathbb{C}[\Lambda]$ is the group algebra. The modular characteristic is $\kappa(V_\Lambda) = r$.

The augmentation removes the vacuum vector. The lattice-sector Epstein sum therefore runs over the punctured lattice $\Lambda \setminus \{0\}$; this set is not a sublattice.

Definition 3.1 (Constrained Epstein zeta function)**.** For a lattice vertex algebra V_Λ of rank r, the *constrained Epstein zeta function* is

$$\varepsilon^c_s(V_\Lambda) := \sum\nolimits'_{\ell\in\Lambda} \|\ell\|^{-2s} = \sum_{\substack{\ell\in\Lambda \\ \ell\neq 0}} \langle \ell, \ell\rangle^{-s},$$

where the prime denotes omission of $\ell = 0$.

This is the Epstein zeta function of the quadratic form $Q(\ell) = \langle \ell, \ell\rangle$ with the zero vector omitted. For a general even lattice, the Epstein zeta function $E_\Lambda(s) = \sum_{\ell\neq 0} Q(\ell)^{-s}$ is a classical object in analytic number theory, studied by Epstein [4, 5] and Siegel [10]. The constraint to Λ^{bar} is the bar-complex restriction: the augmentation ideal removes the vacuum, and the surviving lattice vectors carry the OPE data.

3.2. Recovery of the Riemann zeta function.

Theorem 3.2 (Rank-1 recovery)**.** *For the even rank-1 lattice A_1 with quadratic form $Q(m) = 2m^2$, the constrained Epstein zeta function is*

$$\varepsilon^c_s(V_{A_1}) = 2 \cdot (2)^{-s} \sum_{m\geq 1} m^{-2s} = 2^{1-s}\,\zeta(2s).$$

With the unit quadratic form $Q(m) = m^2$ one obtains $2\zeta(2s)$. The rank-one Narain momentum-winding lattice at radius R gives $2(R^{2s} + R^{-2s})\zeta(2s)$, hence $4\zeta(2s)$ at $R = 1$.

Proof. At rank 1 with $\Lambda = \mathbb{Z}$ and quadratic form $Q(m) = m^2$ (unit radius), the nonzero lattice vectors are $\pm 1, \pm 2, \ldots$, each contributing $|m|^{-2s}$. By symmetry $m \mapsto -m$:

$$\varepsilon^c_s(V_{\mathbb{Z}}) = 2\sum_{m=1}^{\infty} m^{-2s} = 2\,\zeta(2s).$$

For the even lattice with $Q(m) = 2m^2$: $\varepsilon^c_s = 2\cdot(2)^{-s}\,\zeta(2s) = 2^{1-s}\zeta(2s)$. The Narain formula counts the two momentum-winding directions. □

Remark 3.3 (Functional equation)*.* The Epstein zeta function of a rank-r lattice Λ satisfies the functional equation

$$\pi^{-s}\,\Gamma(s)\,E_\Lambda(s) = |\det G|^{-1/2}\,\pi^{-(r/2-s)}\,\Gamma(r/2-s)\,E_{\Lambda^*}(r/2-s),$$

where G is the Gram matrix of Λ and Λ^* is the dual lattice. For the self-dual lattice $\Lambda = \Lambda^*$ at rank 1, this reduces to the functional equation of $\zeta(2s)$. The constrained Epstein zeta inherits this functional equation from the standard Epstein zeta.

3.3. Higher-rank decomposition.

At rank $r \geq 2$, the Epstein zeta function is no longer a product of Riemann zeta functions. The Hecke decomposition provides the correct generalisation.

Proposition 3.4 (Theta-series Hecke decomposition)**.** *For a lattice vertex algebra V_Λ of rank r with Gram matrix G, the completed constrained Epstein zeta is the Mellin transform of $\Theta_\Lambda - 1$. Decomposing the theta series into a Hecke eigenbasis gives*

$$\pi^{-s}\Gamma(s)\varepsilon^c_s(V_\Lambda) = \sum_j c_j\,\Lambda(s, f_j),$$

where the f_j range over Eisenstein and cuspidal Hecke eigenforms occurring in the modular-form space generated by Θ_Λ, and $\Lambda(s, f_j)$ are their completed L-functions.

Proof. The theta series Θ_Λ is a modular form of weight $r/2$ for the appropriate congruence subgroup. The finite-dimensional modular-form space containing it decomposes into Eisenstein and cuspidal Hecke eigenspaces. The Mellin transform of $\Theta_\Lambda(iy) - 1$ is the Epstein zeta integral, and Mellin transform sends each Hecke eigenform summand to its completed L-function. $\square$

Remark 3.5 (Lattice theta function). The lattice theta function $\Theta_\Lambda(\tau) = \sum_{\ell \in \Lambda} q^{Q(\ell)/2}$ is a modular form for a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$, and the constrained Epstein zeta is its Mellin transform:

$$\pi^{-s}\Gamma(s)\, \varepsilon_s^c(V_\Lambda) = \int_0^\infty (\Theta_\Lambda(iy) - 1)\, y^{s-1}\, dy.$$

The subtraction of 1 removes the zero vector.

The Niemeier lattice vertex algebras V_Λ at rank 24 have modular characteristic $\kappa = 24$, independent of the root system. The lattice theta function discriminates between the 24 Niemeier lattices; the shadow tower, which depends only on κ, does not. The constrained Epstein zeta function $\varepsilon_s^c(V_\Lambda)$ carries this finer arithmetic.

4. Depth decomposition of the shadow L-function

4.1. **The shadow L-function.** The shadow obstruction tower $\{S_r(\mathcal{A})\}_{r\geq 2}$ is the sequence of degree-r projections of the Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$, with $S_2 = \kappa(\mathcal{A})$.

Definition 4.1 (Shadow L-function)**.** For a chirally Koszul algebra $\mathcal{A}$, the *shadow L-function* is the Dirichlet series

$$L_{\mathcal{A}}^{\mathrm{sh}}(s) := \sum_{r\geq 2} S_r(\mathcal{A})\, r^{-s}.$$

Example 4.2 (Shadow L-functions by family)**.** The four depth classes produce qualitatively different L-functions.

(a) *Class G* (Heisenberg, $r_{\max} = 2$): $S_r = 0$ for $r \geq 3$, so $L_{\mathcal{H}_k}^{\mathrm{sh}}(s) = k \cdot 2^{-s}$. Class G gives a one-term Dirichlet polynomial.

(b) *Class L* (affine Kac–Moody, $r_{\max} = 3$): $L_{V_k(\mathfrak{g})}^{\mathrm{sh}}(s) = \kappa \cdot 2^{-s} + S_3 \cdot 3^{-s}$. Class L gives a two-term Dirichlet polynomial.

(c) *Class C* ($\beta\gamma$, $r_{\max} = 4$): Three terms; the quartic shadow coefficient S_4 controls the discriminant $\Delta = 8\kappa S_4$.

(d) *Class M* (Virasoro, $\mathcal{W}_N$; $r_{\max} = \infty$): The full Dirichlet series is infinite. For Virasoro: $S_4(\mathrm{Vir}_c) = 10/[c(5c + 22)]$.

4.2. **Analytic properties.**

Proposition 4.3 (Elementary convergence of $L_{\mathcal{A}}^{\mathrm{sh}}$)**.** *For a class-M algebra with shadow coefficients $S_r = O(r^{-\alpha})$ for some $\alpha > 0$, the shadow Dirichlet series $L_{\mathcal{A}}^{\mathrm{sh}}(s)$ converges absolutely for $\Re(s) > 1 - \alpha$.*

Proof. Absolute convergence follows from comparison with $\sum r^{-\alpha-\Re(s)}$. $\square$

No Euler product or Selberg-class structure is part of the definition of $L_{\mathcal{A}}^{\mathrm{sh}}$. Meromorphic continuation, poles, and special values require a separate admissible generating-function theorem for the full shadow tower.

4.3. **Depth stratification.** The four depth classes $\mathbf{G}/\mathbf{L}/\mathbf{C}/\mathbf{M}$ induce a filtration on the space of shadow L-functions.

Definition 4.4 (Depth filtration)**.** The *depth-d truncation* of the shadow L-function is

$$L_{\mathcal{A}}^{\mathrm{sh},\leq d}(s) := \sum_{r=2}^{d} S_r(\mathcal{A})\, r^{-s}.$$

The *depth defect* is $\delta L_{\mathcal{A}}^{\mathrm{sh}}(s) := L_{\mathcal{A}}^{\mathrm{sh}}(s) - L_{\mathcal{A}}^{\mathrm{sh},\leq d}(s) = \sum_{r>d} S_r\, r^{-s}$.

For class G: $\delta L_{\mathcal{A}}^{\mathrm{sh},\leq 2} = 0$ (no defect). For class L: $\delta L_{\mathcal{A}}^{\mathrm{sh},\leq 3} = 0$. For class C: $\delta L_{\mathcal{A}}^{\mathrm{sh},\leq 4} = 0$. For class M: $\delta L_{\mathcal{A}}^{\mathrm{sh},\leq d} \neq 0$ for all finite d; the defect is genuinely infinite.

The scalar discriminant $\Delta(\mathcal{A}) = 8\kappa(\mathcal{A})S_4(\mathcal{A})$ detects the quartic/contact obstruction. It does not distinguish class C from class M. The coarse criteria are:

$$\mathbf{G}:\ S_3 = S_4 = 0, \qquad \mathbf{L}:\ S_3 \neq 0,\ S_4 = 0, \qquad \mathbf{C}:\ S_3 = 0,\ S_4 \neq 0,\ S_{r\geq 5} = 0,$$

while class $\mathbf{M}$ is nonterminating.

5. Arithmetic duality of Bernoulli primes and characteristic primes

This section anchors the arithmetic duality between the Verlinde polynomial family $\{Z_g(k)\}$ and the shadow tower $\{S_r(\mathrm{Vir}_c)\}$. The comparison uses Bernoulli numerators on the Verlinde side and cleared weighted Riccati numerators on the Virasoro shadow side.

5.1. The duality.

Proposition 5.1 (Finite Z_g–S_r arithmetic separation)**.** *The leading Kummer-irregular primes* $\{691, 3617\}$ *are present on* Z_g *through the Hurwitz–Bernoulli leading term* B_{2g-2} *and are absent from the cleared weighted Virasoro shadow numerators* $\widehat{N_r}$ *through* $r = 11$. *Explicitly:*

(i) $B_{12} = -691/2730$ *contributes* $691 \mid \mathrm{num}(Z_7)$.
(ii) $B_{16} = -3617/510$ *contributes* $3617 \mid \mathrm{num}(Z_9)$.
(iii) $\{691, 3617\} \cap \mathrm{supp}(\widehat{N_r}) = \varnothing$ *for* $2 \leq r \leq 11$ *in the computed weighted Riccati range.*

Proof. The first two assertions are the displayed Bernoulli values. The third assertion is the finite factorisation of the cleared weighted Riccati numerators $\widehat{N_r}$ for $2 \leq r \leq 11$. □

The finite comparison is sharp only for the displayed verification window. The prime 2111 appears in the Riccati numerator factorisation $29554 = 2 \cdot 7 \cdot 2111$, but it is not witnessed on the Verlinde leading term for $g \leq 9$ because $2111 \nmid \mathrm{num}(B_{2m})$ for $2m \leq 16$. This is not an ordering statement about Kummer-irregular primes.

5.2. **Riccati-arithmetic characteristic primes.** The cleared weighted numerators $\widehat{N_r}(c)$ through $r = 11$ contain the Riccati-propagated prime factors $\{61, 193, 2111, 16657, 3988097\}$. These primes occur in the integer coefficients of the primitive-cleared polynomials obtained from the Virasoro shadow recursion $H(t)^2 = t^4 Q_L(t)$.

Remark 5.2 (Kummer and Riccati-prime separation)*.* In the computed range $2 \leq r \leq 11$, the primes $\{1423, 23, 43, 419\}$ are absent from the Kummer-irregular list and occur only on the Riccati side. The prime 3067 is not witnessed as Kummer-irregular at the current verification depth. The structural distinction is geometric: Kummer-arithmetic primes track B_{2g-2} on the Verlinde side, while Riccati-arithmetic primes track the Kac determinant on the shadow side.

Remark 5.3 (Complement: S_r-characteristic primes, absent from leading Z_g). The characteristic primes $\{61, 193, 2111, 16657, 3988097\}$ occur in $\mathrm{num}(S_r(\mathrm{Vir}_c))$ for suitable r but do *not* divide $\mathrm{num}(B_{2g-2})$ at $g \leq 9$. Their emergence is Riccati-seed-driven: they occur in the integer coefficients produced by the shadow recursion and are invisible to the Hurwitz–Bernoulli leading term on Z_g in the displayed range.

6. Verlinde polynomials from ordered chiral homology

6.1. **Recovery of the Verlinde formula.** At integer level $k \geq 1$, the quantum group parameter $q = e^{2\pi i/(k+2)}$ is a root of unity, the representation category of $\widehat{\mathfrak{sl}}_2$ at level k truncates to $k+1$ integrable modules, and the ordered chiral homology at each genus computes a finite-dimensional invariant. Write $n = k+2$ for the shifted level.

Proposition 6.1 (Verlinde formula from ordered chiral homology). *Let $k \geq 1$ be a positive integer, and let $S_{jl} = \sqrt{2/n}\,\sin\bigl(\pi(j+1)(l+1)/n\bigr)$ be the modular S-matrix for $\widehat{\mathfrak{sl}}_2$ at level k, where $j, l \in \{0, 1, \dots, k\}$.*

(i) (Genus 0.) $\dim H^0\bigl(\mathcal{M}_{0,0}, V_k(\mathfrak{sl}_2)\bigr) = \sum_{j=0}^{k} S_{0j}^2 = 1.$

(ii) (Genus 1.) $\dim H^0\bigl(\mathcal{M}_{1,0}, V_k(\mathfrak{sl}_2)\bigr) = \sum_{j=0}^{k} S_{0j}^0 = k+1.$

(iii) (General genus.) $Z_g(k) := \dim H^0\bigl(\mathcal{M}_{g,0}, V_k(\mathfrak{sl}_2)\bigr) = \sum_{j=0}^{k} S_{0j}^{2-2g}.$

(iv) (Handle attachment.) $Z_{g+1} = \sum_{j=0}^{k} S_{0j}^{-2} \cdot S_{0j}^{2-2g} = \sum_{j=0}^{k} S_{0j}^{2-2(g+1)}.$

(v) (Separating factorization.) *Under* $\Sigma_g \rightsquigarrow \Sigma_{g_1} \cup \Sigma_{g_2}$ *with* $g = g_1 + g_2$*:* $Z_g = \sum_{j=0}^{k} S_{0j}^{2-2g_1} \cdot S_{0j}^{2-2g_2} \cdot S_{0j}^{-2}.$

Proof. The identity $\sum_j S_{0j}^2 = 1$ is the $(0,0)$-entry of $S \cdot S^T = I$, following from the orthogonality relation $\sum_{j=0}^{k} \sin(\pi(j+1)a/n)\sin(\pi(j+1)b/n) = n\,\delta_{ab}/2$. Since $S_{0j}^0 = 1$ for all j, $Z_1 = k+1$. The KZB local system at integrable level decomposes into $k+1$ channels labeled by integrable representations; each channel j contributes S_{0j}^{2-2g} to the partition function. Handle attachment and separating factorization are the same diagonal channel computation under sewing. □

The first values are:

k	Z_0	Z_1	Z_2	Z_3	Z_4
1	1	2	4	8	16
2	1	3	10	36	136
3	1	4	20	120	800
4	1	5	35	329	3,611
5	1	6	56	784	13,328

Remark 6.2 (Two genus invariants). The shadow coefficients F_g and the Verlinde dimensions Z_g are different invariants of the same algebra. The shadow coefficient $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ varies continuously with the level and is a small rational number. For $V_1(\mathfrak{sl}_2)$ one has $\kappa = 9/4$, hence $F_2 = 7/2560$. The Verlinde dimension Z_g is a positive integer defined only at integer level, with $Z_2 = 4$ at $k = 1$.

6.2. **The Verlinde polynomial family.** The Verlinde dimensions at each genus are polynomials in $n = k+2$.

Theorem 6.3 (Verlinde polynomial family). *For $\widehat{\mathfrak{sl}}_2$ at level $k \geq 1$ and genus $g \geq 2$, the Verlinde dimension $Z_g(k) = \sum_{j=0}^{k} S_{0j}^{2-2g}$ is a polynomial $P_g(n)$ of degree $3(g-1)$ in $n = k+2$ with the following structure.*

(i) (Universal factor.) *For $g \geq 2$,*

$$P_g(n) = n^{g-1}(n^2-1) \cdot R_{g-2}(n^2), \tag{6.1}$$

where R_{g-2} is a polynomial of degree $g-2$ in n^2 with rational coefficients and positive leading term.

(ii) (Explicit formulas through $g = 6$.)

$$P_2(n) = \frac{n(n^2-1)}{6}, \tag{6.2}$$
$$P_3(n) = \frac{n^2(n^2-1)(n^2+11)}{180},$$
$$P_4(n) = \frac{n^3(n^2-1)(2n^4+23n^2+191)}{7560},$$
$$P_5(n) = \frac{n^4(n^2-1)(n^2+11)(3n^4+10n^2+227)}{226800},$$
$$P_6(n) = \frac{n^5(n^2-1)(2n^8+35n^6+321n^4+2125n^2+14797)}{2993760}. \tag{6.3}$$

(iii) (Leading asymptotics.) *The leading coefficient as $n \to \infty$ is*

$$P_g(n) \sim \frac{\zeta(2g-2)}{2^{g-2}\,\pi^{2g-2}}\, n^{3(g-1)} \qquad (n \to \infty).$$

(iv) (Rational generating function.) *At fixed n,*

$$G_n(x) := \sum_{g \geq 1} Z_g(k)\, x^{g-1} = \sum_{j=1}^{n-1} \frac{1}{1 - a_j x}, \qquad a_j = \frac{n}{2\sin^2(\pi j/n)}.$$

Proof. The cosecant-power representation $Z_g(k) = (n/2)^{g-1} \sum_{j=1}^{n-1} \csc^{2g-2}(\pi j/n)$ follows from $S_{0j} = \sqrt{2/n}\,\sin(\pi j/n)$. The cosecant power sum $\sigma_m(n) = \sum_{j=1}^{n-1} \csc^{2m}(\pi j/n)$ is a polynomial of degree $2m$ in n with rational coefficients: a classical result of Euler, with leading coefficient $\sigma_m(n) \sim 2\zeta(2m)\pi^{-2m} n^{2m}$. Parts (i)–(iii) follow. Part (iv): the geometric series $\sum_{g\geq 1} a_j^{g-1} x^{g-1} = 1/(1 - a_j x)$ applied term by term gives the rational generating function. The explicit formulas (6.2)–(6.3) are obtained by polynomial interpolation and verified by direct S-matrix summation. $\square$

Remark 6.4 (Zeta values in the Verlinde tower). The leading coefficient $[n^{3g-3}]P_g = \zeta(2g-2)/(2^{g-2}\pi^{2g-2})$ connects the Verlinde tower to zeta values at even positive integers:

$$\frac{\zeta(2)}{2^0\,\pi^2} = \frac{1}{6}, \quad \frac{\zeta(4)}{2\,\pi^4} = \frac{1}{180}, \quad \frac{\zeta(6)}{4\,\pi^6} = \frac{1}{3780}, \quad \frac{\zeta(8)}{8\,\pi^8} = \frac{1}{75600}.$$

The displayed denominators are the reduced leading-coefficient denominators of P_2, P_3, P_4, P_5.

Remark 6.5 (Structural observations). Three features of the polynomial family (6.1):

(a) The factor $n^2 - 1 = (k+1)(k+3)$ vanishes at $k = -1$ and $k = -3$, reflecting the boundary of the integrable range. The factor n^{g-1} vanishes at $n = 0$ ($k = -2$, the critical level), where the Verlinde formula degenerates.

(b) $P_2(n) = \binom{n+1}{3}$ is the unique genus at which Z_g is a single binomial coefficient (the tetrahedral numbers).

(c) The factor $(n^2 + 11)$ appears in both P_3 and P_5, suggesting a recurrence in the polynomials R_{g-2}.

7. The Böcherer bridge: genus-2 Siegel modular forms

7.1. From the Maurer–Cartan element to Siegel modular forms. For lattice VOAs and rational conformal-block systems satisfying the analytic sewing hypotheses, the genus-g theta or conformal-block amplitudes transform under $\mathrm{Sp}(2g, \mathbb{Z})$. The genus-2 lattice case is the first place where the bar-complex theta component meets the arithmetic of Siegel modular forms.

The genus-2 bar complex on Σ_2 with Siegel period matrix $\Omega \in \mathbb{H}_2 = \{Z \in M_2(\mathbb{C}) : Z = Z^t, \ \mathrm{Im}\, Z > 0\}$ involves the prime form $E(z, w)$ on Σ_2 and the matrix $(\mathrm{Im}\,\Omega)^{-1}$ (a 2×2 matrix, not a scalar; at $g = 1$ this reduces to $1/\mathrm{Im}(\tau)$). The bar propagator is $\eta_{ij}^{(2)} = d \log E(z_i, z_j)$, weight 1.

7.2. The Hecke decomposition at genus 2. The Hecke operators T_p act on the space $M_{\mathcal{A}}$ of activated amplitudes, decomposing $M_{\mathcal{A}} = \bigoplus_\chi M_\chi$ into generalised eigenspaces. The *arithmetic packet connection*

$$\nabla_{\mathcal{A}}^{\mathrm{arith}} = d - \bigoplus_{\chi} \frac{\Lambda'_\chi(s)}{\Lambda_\chi(s)} (\mathrm{Id}_{M_\chi} + N_\chi)\, ds$$

is a flat meromorphic connection on $M_{\mathcal{A}} \otimes \mathcal{O}_{\mathbb{C}}$ over the spectral s-line, where $\Lambda_\chi(s)$ is the completed L-function of the eigenform and N_χ is the nilpotent part. The singular divisor depends only on the semisimple part $M_{\mathcal{A}}^{\mathrm{ss}}$: the arithmetic is semisimple, the homotopy defect is unipotent.

7.3. The Saito–Kurokawa lift and the Böcherer theorem. For even unimodular lattice vertex algebras, the genus-2 amplitude is a Siegel modular form for $\mathrm{Sp}(4, \mathbb{Z})$, and Böcherer's theorem relates its Fourier coefficients to central L-values in the Saito–Kurokawa case.

Theorem 7.1 (Leech χ_{12} projection)**.** *For the Leech lattice vertex algebra, the genus-2 Siegel theta component has nonzero projection to the one-dimensional cusp space $S_{12}(\mathrm{Sp}(4, \mathbb{Z})) = \mathbb{C}\chi_{12}$. The cusp form is*

$$\chi_{12} = \mathrm{SK}(f_{22}),$$

where $f_{22} = \Delta E_{10}$ is the weight-22 Hecke eigenform. For fundamental $D < 0$, the Saito–Kurokawa/Böcherer coefficient $B(D; \chi_{12})$ equals the half-integral-weight coefficient $c(|D|)$, and Waldspurger gives

$$|c(|D|)|^2 \ \propto \ L(11, f_{22} \times \chi_D)\, |D|^{10}.$$

Proof. The cusp-space dimension and the identity $\chi_{12} = \mathrm{SK}(f_{22})$ are the Igusa–Saito–Kurokawa input. The nonzero projection is computed from the Igusa relation and the genus-2 Leech representation numbers; the coefficient is $c_2 \approx -1.918 \times 10^{-6}$ with zero residual in the exact rational computation. The last assertion is Waldspurger's theorem applied to the half-integral-weight form attached to the Saito–Kurokawa lift; Furusawa–Morimoto [6] give the refined Böcherer form. □

Remark 7.2 (Arithmetic input). Deligne's theorem supplies the Ramanujan bound for the Hecke eigenforms appearing in the genus-2 decomposition. The Maurer–Cartan equation supplies the chiral source of the theta component; it does not replace the arithmetic proof of the bound.

8. The constrained Epstein zeta and the Dirichlet–sewing lift

8.1. The Dirichlet–sewing lift. The genus-1 connected free energy $F_1^{\mathrm{conn}}(\mathcal{A};q) = -\log\det(1-K_q)$ encodes connected sewing amplitudes $a_{\mathcal{A}}(N)$ as Fourier coefficients. The *Dirichlet–sewing lift* is

$$S_{\mathcal{A}}(u) = \sum_{N\geq 1} a_{\mathcal{A}}(N)\, N^{-u} = \frac{(2\pi)^u}{\Gamma(u)} \int_0^\infty F_{\mathcal{A}}^{\mathrm{conn}}(e^{-2\pi y})\, y^{u-1}\, dy.$$

This is a Dirichlet series whose analytic structure is controlled by $\Theta_{\mathcal{A}}$.

Example 8.1 (Sewing Dirichlet series by family)**.**

$$\begin{aligned} S_{\mathcal{H}}(u) &= \zeta(u)\,\zeta(u+1), \\ S_{V_k(\mathfrak{g})}(u) &= \dim(\mathfrak{g})\cdot\zeta(u)\,\zeta(u+1), \\ S_{\mathrm{Vir}}(u) &= \zeta(u+1)\,(\zeta(u)-1), \\ S_{\mathcal{W}_N}(u) &= \zeta(u+1)\Big((N-1)\,\zeta(u) - \sum_{j=1}^{N-1} H_j(u)\Big), \end{aligned}$$

where $H_j(u) = \sum_{m=1}^{j} m^{-u}$ are partial harmonic sums.

Remark 8.2 (The Heisenberg base point)*.* The Heisenberg sewing Dirichlet series $S_{\mathcal{H}}(u) = \zeta(u)\,\zeta(u+1)$ is a product of two shifted Riemann zeta functions, connecting the sewing amplitudes to the convolution structure of divisor sums. Every other family's sewing series decomposes as the Heisenberg core plus a defect: the affine Kac–Moody series is $\dim(\mathfrak{g})$ copies of Heisenberg, the Virasoro series subtracts the constant term 1, and $\mathcal{W}_N$ has $N-1$ Heisenberg copies minus partial harmonic corrections.

8.2. The three-tier Euler–Koszul classification. The *Euler defect* $D_{\mathcal{A}}(u) := S_{\mathcal{A}}(u)/(|W(\mathcal{A})|\,\zeta(u)\,\zeta(u+1))$, where $|W(\mathcal{A})|$ is the number of strong generators, classifies chiral algebras into three tiers.

Tier	**Condition**	**Families**
Exact Euler–Koszul	$D_{\mathcal{A}}(u) \equiv 1$	$\mathcal{H}, V_k(\mathfrak{g}), \beta\gamma$
Finitely defective	$1 - D_{\mathcal{A}}(u)$ finite harmonic ratio	Virasoro, $\mathcal{W}_N$
Non-Euler–Koszul	$S_{\mathcal{A}}$ decomposes into Hecke L-functions	Lattice VOAs with $h(D)\geq 2$

The Euler–Koszul class $\mathrm{ek}(\mathcal{A}) = \max_i(w_i - 1)$, where w_i are the conformal weights of the strong generators, and the shadow depth $r_{\max}(\mathcal{A})$ are independent invariants: character arithmetic and OPE interaction are logically independent axes of complexity.

8.3. The two-variable L-object. Under the diagonal weight-multiset sewing hypotheses, Rankin–Selberg unfolding against $E^*(\tau,s)$ gives

$$L_{\mathcal{A}}(s,u) = \int_{\mathcal{F}^{\mathrm{reg}}} F_{\mathcal{A}}^{\mathrm{conn}}(\tau)\, E^*(\tau,s)\, y^{u-2}\, dx\, dy,$$

where $\mathcal{F}^{\mathrm{reg}}$ is the regularised fundamental domain. The same hypotheses give the one-variable reduction

$$L_{\mathcal{A}}(s,u) = \frac{\Gamma(v)}{(2\pi)^v}\, S_{\mathcal{A}}(v), \qquad v = s+u-1.$$

The scattering poles of E^* at $s=\rho/2$ for nontrivial zeros ρ of ζ become structural obstructions to an Euler product for the two-variable object.

8.4. **Constrained Epstein zeta at higher rank.** For a lattice VOA V_Λ, the algebraic shadow depth and the arithmetic Epstein depth are separate invariants. In the diagonal sector the constituent-centre count has the form

$$d = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}},$$

where d_{arith} comes from the automorphic components of the theta series and d_{alg} records algebraic finite-defect factors. Lattice VOAs remain Gaussian in the shadow depth sense; the Epstein arithmetic depth records the lattice rank and theta decomposition.

Remark 8.3 (The finite Miura defect). The principal $\mathcal{W}_N$ character is $\chi_{\mathcal{W}_N}(q) = \prod_{n\geq 2}(1 - q^n)^{-\min(n-1,N-1)}$; the Heisenberg character is $\chi_{\mathcal{H}}(q) = \prod_{n\geq 1}(1-q^n)^{-1}$. The ratio

$$\chi_{\mathcal{W}_N}(q) = \chi_{\mathcal{H}}(q)^{N-1} \cdot \prod_{m=1}^{N-1}(1-q^m)^{N-m}$$

is a finite product: the *Miura defect*. All genus-1 arithmetic content in $\mathcal{W}_N$ beyond Heisenberg is carried by this finite product. The prime-side Dirichlet analogue is $D^{\mathrm{prime}}_{\mathcal{W}_N}(u) = -\zeta(u+1)\sum_{m=1}^{N-1}(N-m)\,m^{-u}$; the packet connection splits into $(N-1)$ Heisenberg copies plus a defect governing modes $m = 1, \ldots, N-1$.

Remark 8.4 (Shadow-moduli resolution). The shadow-moduli map $t\colon \mathfrak{H} \to \mathbb{C}$ solves $\prod_j(1-\lambda_j\, t(q))^{c_j} = \det(1 - K_q^{\mathrm{conn}})$ with spectral atoms λ_j and multiplicities c_j. The connected free energy factorises as $F_1^{\mathrm{conn}}(q) = G_{\mathcal{A}}(t(q))$ with

$$G_{\mathcal{A}}(t) = \int \log(1-\lambda t)\, d\rho_{\mathcal{A}}(\lambda),$$

determined by the spectral measure $\rho_{\mathcal{A}}$ alone. The moments $\mu_r = \int \lambda^r\, d\rho_{\mathcal{A}}$ satisfy the MC recursion degree by degree: the bar-complex differential equations control the spectral resolution of the genus-1 amplitude.

References

[1] A. Beilinson and V. Drinfeld, *Chiral algebras*, Amer. Math. Soc. Colloq. Publ., vol. 51, AMS, 2004.

[2] S. Böcherer, Über die Fourier-Jacobi-Entwicklung Siegelscher Eisensteinreihen II, *Math. Z.* **189** (1985), 81–110.

[3] K. Costello and O. Gwilliam, *Factorization algebras in quantum field theory*, Vol. 1 and 2, Cambridge Univ. Press, 2017 and 2021.

[4] P. Epstein, Zur Theorie allgemeiner Zetafunctionen, *Math. Ann.* **56** (1903), 615–644.

[5] P. Epstein, Zur Theorie allgemeiner Zetafunctionen. II, *Math. Ann.* **63** (1907), 205–216.

[6] M. Furusawa and K. Morimoto, Refined global Gross–Prasad conjecture on special Bessel periods and Böcherer's conjecture, *J. Eur. Math. Soc. (JEMS)* **23** (2021), 1295–1331.

[7] H. Iwaniec and E. Kowalski, *Analytic number theory*, American Mathematical Society Colloquium Publications, 53. American Mathematical Society, Providence, RI, 2004.

[8] E. Frenkel and D. Ben-Zvi, *Vertex algebras and algebraic curves*, 2nd ed., Math. Surv. Monogr., vol. 88, AMS, 2004.

[9] V. Kac, *Vertex algebras for beginners*, 2nd ed., Univ. Lecture Ser., vol. 10, AMS, 1998.

[10] C. L. Siegel, Indefinite quadratische Formen und Funktionentheorie. I, II, *Math. Ann.* **124** (1951/52), 17–54, 364–387.

[11] A. Tsuchiya, K. Ueno, and Y. Yamada, Conformal field theory on universal family of stable curves with gauge symmetries, *Adv. Stud. Pure Math.* **19** (1989), 459–566.

[12] E. Verlinde, Fusion rules and modular transformations in 2D conformal field theory, *Nuclear Phys. B* **300** (1988), 360–376.

[13] D. Zagier, Elliptic modular forms and their applications, in *The 1-2-3 of modular forms*, Springer, 2008, pp. 1–103.

[14] Y. Zhu, Modular invariance of characters of vertex operator algebras, *J. Amer. Math. Soc.* **9** (1996), 237–302.

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